

실습 가이드: OpenFOAM cavity flow

2026 한국연소학회 가을 Tutorial
Next-Gen Combustion

Cavity Flow

• Tutorial 1: cavity flow

- Describe how to pre-process, run and post-process a case involving isothermal, incompressible flow
- **Lid-driven cavity flow** (see Fig. 1)
 - 2D square domain
 - Top wall moves in the x-direction at 1 m/s and other 3 walls are stationary.
 - incompressibleFluid solver module:
for laminar, isothermal, incompressible flow.
 - General procedure for CFD simulations.
 - Pre-processing (e.g., mesh generation)
 - Run the simulation
 - Post-processing

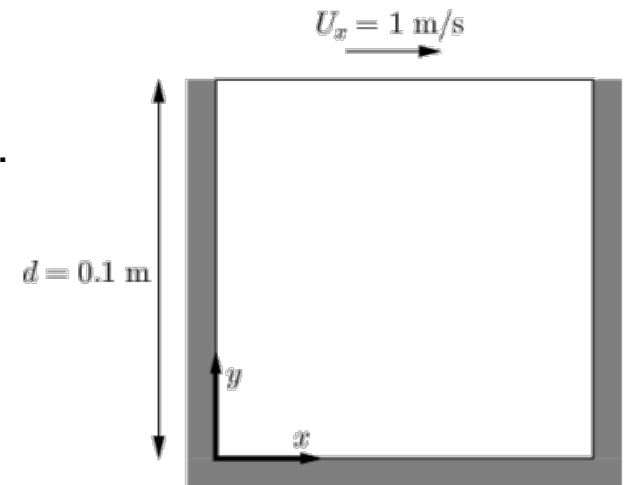
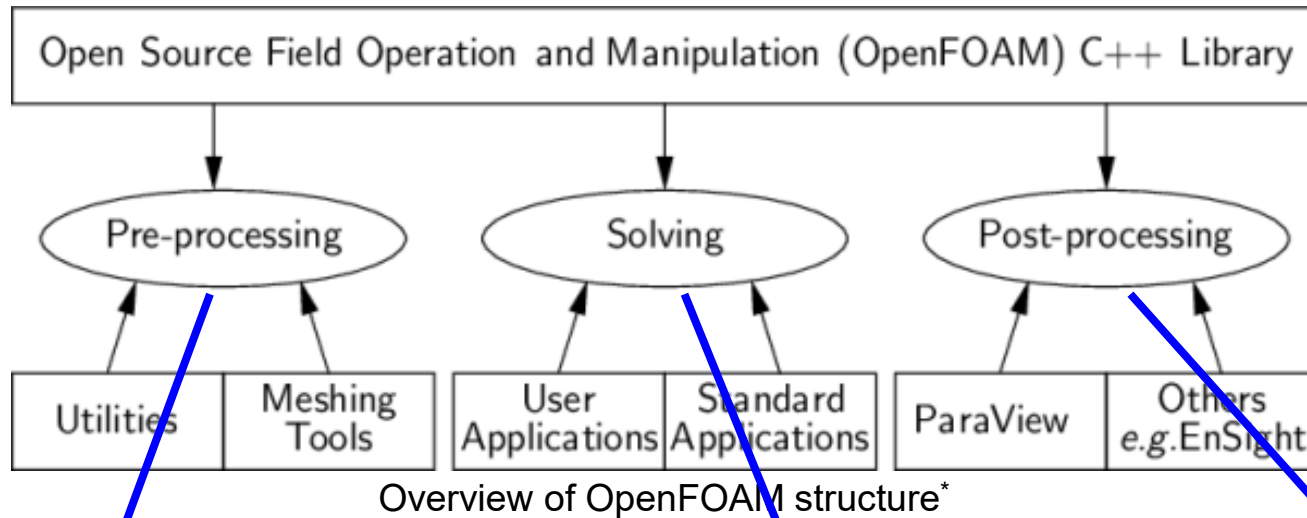


Figure 1. Geometry of the lid driven cavity

Cavity Flow

● Overall structure of OpenFOAM



OpenFOAM provide the **standard mesh generation tool** (e.g., BlockMesh)

Utility for conversion of generated mesh by third-party products (e.g., Ansys Fluent → FluentMeshToFoam)

Various **pre-built solvers** (e.g., multiComponentFluid)

Users can **extend or modify** the libraries and solvers of standard OpenFOAM.

Compatible with third-party **visualization tools** (e.g., Paraview, tecplot, EnSight, etc.,)

Cavity Flow

- **Governing equation of incompressibleFluid**

- A solver module for transient, incompressible, isothermal laminar flow simulation
- For incompressible flow, the pressure is expressed as kinematic pressure $\frac{p}{\rho}$
- Governing equations:

$$\nabla \cdot \mathbf{u} = 0$$

$$\frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot (\mathbf{u}\mathbf{u}) - \nabla \cdot (\nu \nabla \mathbf{u}) = -\nabla \frac{p}{\rho}$$

Cavity Flow

- Pre-processing

- Get the tutorial files
- The recommended location is your OpenFOAM run directory

Command Line

```
cd $FOAM_RUN
```

```
ghkrk@20225263-p01:~$ ll
total 68
drwxr-x--- 7 ghkrk ghkrk 4096 Oct  1 12:50 ./
drwxr-xr-x 3 root  root  4096 Sep 30 17:18 ../
drwxr-xr-x 3 ghkrk ghkrk 4096 Sep 30 17:31 .OpenFOAM/
-rw----- 1 ghkrk ghkrk  436 Oct  1 12:52 .bash_history
-rw-r--r-- 1 ghkrk ghkrk   220 Sep 30 17:18 .bash_logout
-rw-r--r-- 1 ghkrk ghkrk 3875 Sep 30 17:30 .bashrc
drwx----- 2 ghkrk ghkrk 4096 Sep 30 17:18 .cache/
drwxr-xr-x 2 ghkrk ghkrk 4096 Oct  1 12:37 .landscape/
-rw-r--r-- 1 ghkrk ghkrk     0 Oct  1 12:37 .motd_shown
-rw-r--r-- 1 ghkrk ghkrk   807 Sep 30 17:18 .profile
-rw-r--r-- 1 ghkrk ghkrk     0 Sep 30 17:18 .sudo_as_admin_successful
-rw----- 1 ghkrk ghkrk 23391 Oct  1 12:50 .viminfo
drwxr-xr-x 3 ghkrk ghkrk 4096 Sep 30 17:29 OpenFOAM/
drwxr-xr-x 2 ghkrk ghkrk 4096 Sep 30 17:42 -p/
ghkrk@20225263-p01:~$ cd $FOAM_RUN
ghkrk@20225263-p01:~/OpenFOAM/ghkrk-12/run$ ll
```

Cavity Flow

- **Pre-processing**

- Get the tutorial files
- The recommended location is your OpenFOAM run directory

Command Line

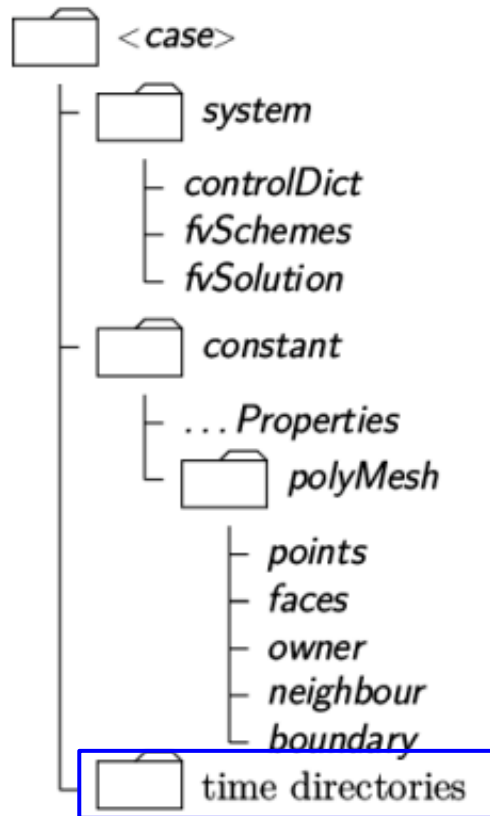
```
git clone https://github.com/CCERL/KOSCO-2026-NextGen-Combustion.git
```

```
cd KOSCO-2026-NextGen-Combustion/Tutorial/01-cavity
```

```
ghkrk@20225263-p01:~/OpenFOAM/ghkrk-12/run$ git clone https://github.com/CCERL/KOSCO-2026-NextGen-Combustion.git
Cloning into 'KOSCO-2026-NextGen-Combustion'...
remote: Enumerating objects: 236, done.
remote: Counting objects: 100% (236/236), done.
remote: Compressing objects: 100% (164/164), done.
remote: Total 236 (delta 126), reused 126 (delta 69), pack-reused 0 (from 0)
Receiving objects: 100% (236/236), 2.16 MiB | 7.83 MiB/s, done.
Resolving deltas: 100% (126/126), done.
ghkrk@20225263-p01:~/OpenFOAM/ghkrk-12/run$ cd KOSCO-2026-NextGen-Combustion/Tutorial/01-cavity
ghkrk@20225263-p01:~/OpenFOAM/ghkrk-12/run/KOSCO-2026-NextGen-Combustion/Tutorial/01-cavity$ ll
total 20
drwxr-xr-x 5 ghkrk ghkrk 4096 Oct  1 12:56 ./
drwxr-xr-x 5 ghkrk ghkrk 4096 Oct  1 12:56 ../
drwxr-xr-x 2 ghkrk ghkrk 4096 Oct  1 12:56 0/
drwxr-xr-x 2 ghkrk ghkrk 4096 Oct  1 12:56 constant/
drwxr-xr-x 2 ghkrk ghkrk 4096 Oct  1 12:56 system/
```

Cavity Flow

• File structure of OpenFOAM cases



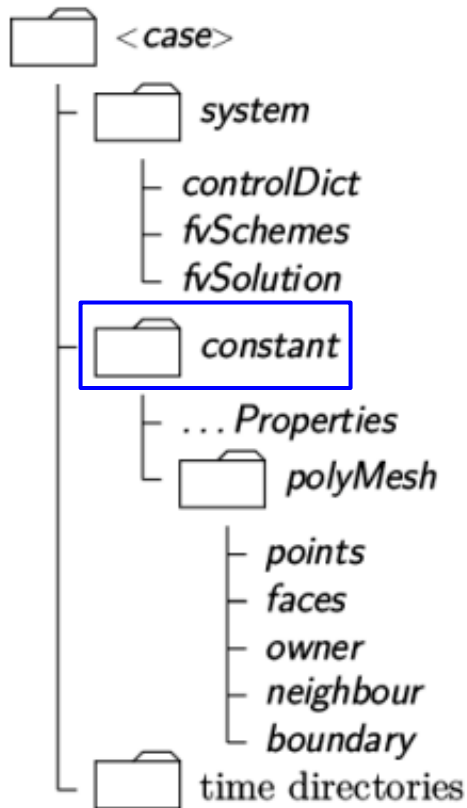
- time directory: contains individual files of data for particular fields (e.g., U and p). Each data files can be either
 - **Initial fields** and **boundary conditions** that the user must specify to define the problem
 - Output file
- The name of each time directory is based on the simulated time at which the data is written.

```
4096 Sep 30 19:18 0
4096 Sep 30 19:18 constant
4096 Sep 30 19:18 system
```

Overview of file structure of OpenFOAM case*

Cavity Flow

• File structure of OpenFOAM cases

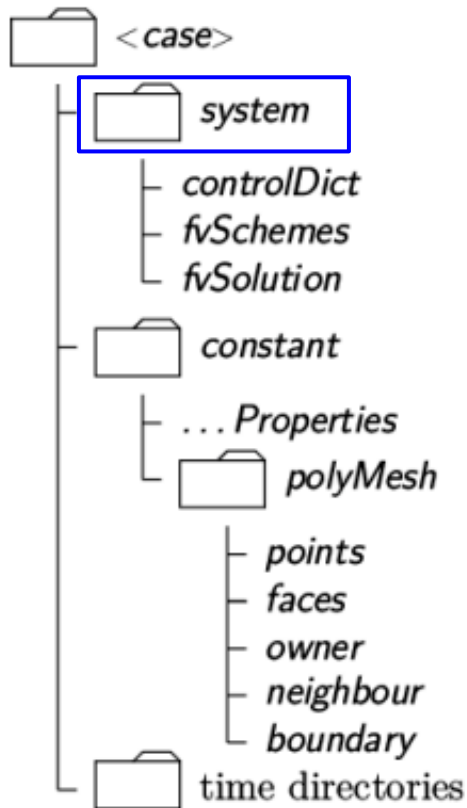


- constant directory: contains
 - polyMesh : Full collection of the mesh files
 - files specifying physical properties for the application concerned
(e.g., momentumTransport, physicalProperties, chemistryProperties)

Overview of file structure of OpenFOAM case*

Cavity Flow

• File structure of OpenFOAM cases



- system directory: contains **at least** the following 3 files:
 - `controlDict`: run control parameters are set including start/end time, time step and parameters for data output.
 - `fvSchemes`: discretization schemes used in the solution
 - `fvSolution`: the equation solvers, tolerances and other algorithm controls are set for the run.

Overview of file structure of OpenFOAM case*

Cavity Flow

- **Steps for CFD simulation using OpenFOAM**

1. Define the problem
2. Find the proper solver in OpenFOAM
3. Make the geometry and mesh
4. Set up the necessary conditions
 - time directory : initial & boundary conditions
 - constant : Physical properties & models & mesh
 - system : simulation control & numerical settings

Pre-processing [

Solving – 5. Run the simulation (start the calculation)

Post-processing – 6. Check the results of the simulation

Cavity Flow

● Preprocessing: Boundary and Initial Conditions

- Every field file consists of *internalField* and *boundayField*.

```

/*-----*- C++ -*-----*/
=====
\\      /  F i e l d      |  OpenFOAM: The Open Source CFD Toolbox
\\      /  O p e r a t i o n  |  Website:  https://openfoam.org
\\      /  A n d      |  Version:  12
\\      /  M a n i p u l a t i o n  |
\\-----/

FoamFile
{
    format      ascii;
    class       volVectorField;
    object      U;
}
// *****

dimensions      [0 1 -1 0 0 0 0];

internalField    uniform (0 0 0);

boundaryField
{
    movingWall
    {
        type      fixedValue;
        value      uniform (1 0 0);
    }

    fixedWalls
    {
        type      noSlip;
    }

    frontAndBack
    {
        type      empty;
    }
}

// *****

```

Physical dimension of the field.

Addition, subtraction and equality are only physically meaningful for properties of the same dimensional units. (As a safeguard against implementing a meaningless operation)

[kg m s K mol A cd].

Cavity Flow

● Preprocessing: Boundary and Initial Conditions

```
/*----- C++ -----*/
//
//  F i e l d
//  O p e r a t i o n
//  A n d
//  M a n i p u l a t i o n
//
//  OpenFOAM: The Open Source CFD Toolbox
//  Website:  https://openfoam.org
//  Version:  12
//
FoamFile
{
    format      ascii;
    class       volVectorField;
    object      U;
}
// ***** //

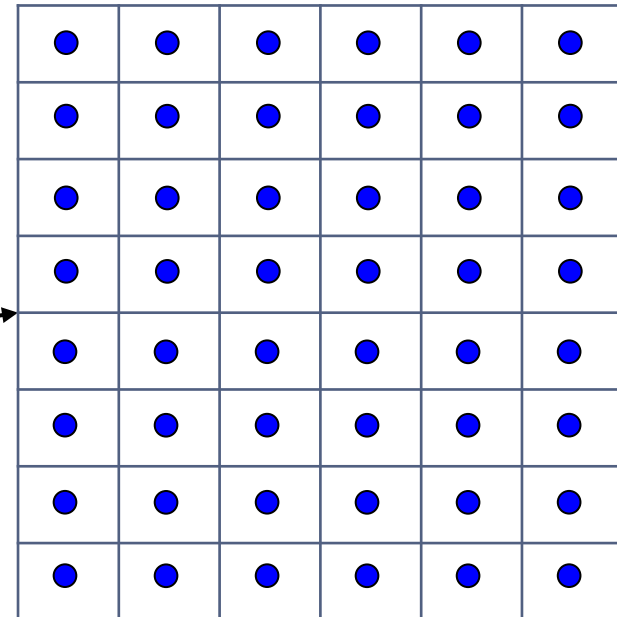
dimensions      [0 1 -1 0 0 0];

internalField    uniform (0 0 0);

boundaryField
{
    movingWall
    {
        type      fixedValue;
        value      uniform (1 0 0);
    }

    fixedWalls
    {
        type      noSlip;
    }

    frontAndBack
    {
        type      empty;
    }
}
// ***** //
```



internalField of U is uniformly set to (0 0 0) m/s
- Initial condition

Cavity Flow

● Preprocessing: Boundary and Initial Conditions

```
/*----- C++ -----*/
//
//  F i e l d
//  O p e r a t i o n
//  A n d
//  M a n i p u l a t i o n
//
//  OpenFOAM: The Open Source CFD Toolbox
//  Website: https://openfoam.org
//  Version: 12
//
FoamFile
{
    format      ascii;
    class       volVectorField;
    object      U;
}
// ***** //

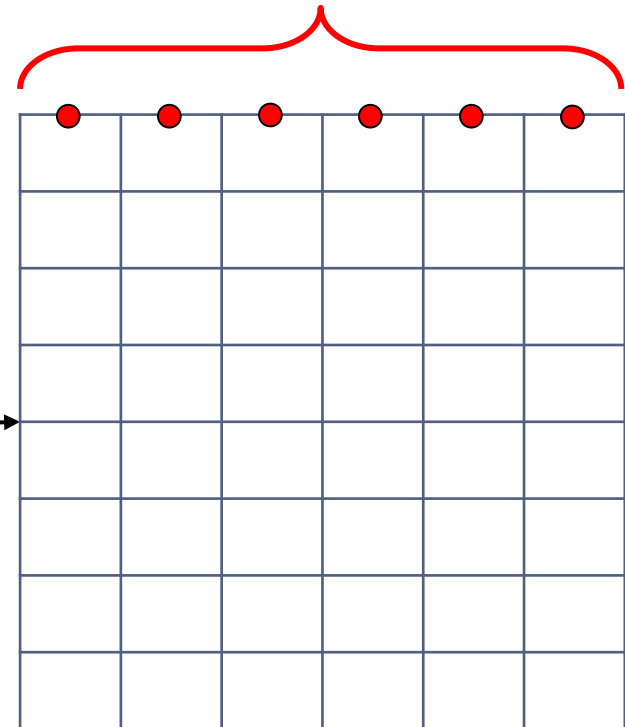
dimensions      [0 1 -1 0 0 0];
internalField    uniform (0 0 0);

boundaryField
{
    movingWall
    {
        type      fixedValue;
        value      uniform (1 0 0);
    }

    fixedWalls
    {
        type      noSlip;
    }

    frontAndBack
    {
        type      empty;
    }
}
// ***** //
```

U at the movingWall
(1 0 0) m/s



Cavity Flow

● Preprocessing: Boundary and Initial Conditions

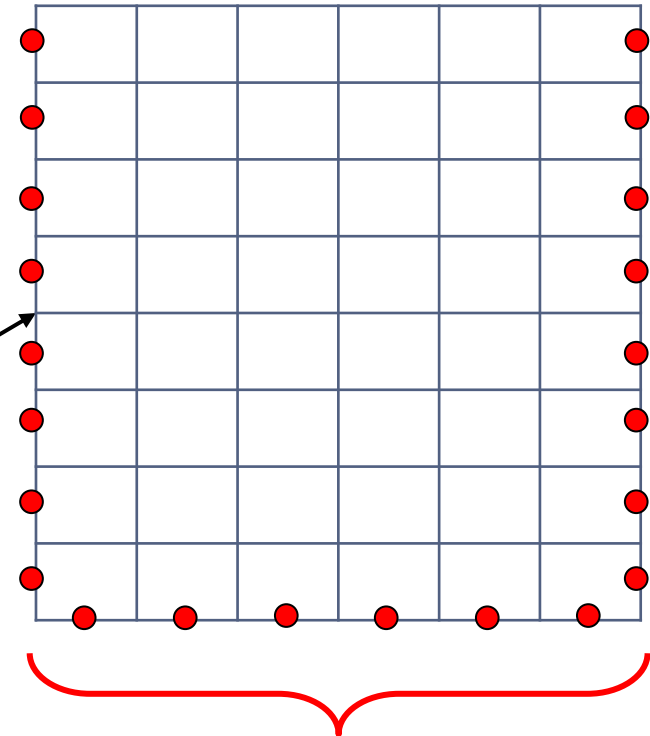
```
/*----- C++ -----*/
//
//  F i e l d
//  O p e r a t i o n
//  A n d
//  M a n i p u l a t i o n
//
//  OpenFOAM: The Open Source CFD Toolbox
//  Website: https://openfoam.org
//  Version: 12
//
FoamFile
{
    format      ascii;
    class       volVectorField;
    object      U;
}
// ***** //

dimensions      [0 1 -1 0 0 0];
internalField    uniform (0 0 0);

boundaryField
{
    movingWall
    {
        type      fixedValue;
        value      uniform (1 0 0);
    }

    fixedWalls
    {
        type      noSlip;
    }

    frontAndBack
    {
        type      empty;
    }
}
// ***** //
```



U at the remaining wall
no slip boundary condition

Cavity Flow

- Preprocessing: Boundary and Initial Conditions

```
/*----- C++ -----*\
\=====
\ \  F i e l d      | | OpenFOAM: The Open Source CFD Toolbox
\ \  O p e r a t i o n | | Website: https://openfoam.org
\ \  A n d             | | Version: 12
\ \  M a n i p u l a t i o n | |
\=====
\*-----*/

FoamFile
{
    format      ascii;
    class       volVectorField;
    object      U;
}

// ***** //

dimensions      [0 1 -1 0 0 0];

internalField    uniform (0 0 0);

boundaryField
{
    movingWall
    {
        type      fixedValue;
        value      uniform (1 0 0);
    }

    fixedWalls
    {
        type      noSlip;
    }

    frontAndBack
    {
        type      empty;
    }
}

// ***** //
```

frontAndBack, which are the faces normal to z, is set to empty

Cavity Flow

● Preprocessing: Boundary and Initial Conditions

```
/*----- C++ -----*/
//
//  F i e l d
//  O p e r a t i o n
//  A n d
//  M a n i p u l a t i o n
//
//  OpenFOAM: The Open Source CFD Toolbox
//  Website: https://openfoam.org
//  Version: 12
//
FoamFile
{
    format      ascii;
    class       volScalarField;
    object      p;
}
// ***** //

dimensions      [0 2 -2 0 0 0];

internalField   uniform 0;

boundaryField
{
    movingWall
    {
        type      zeroGradient;
    }

    fixedWalls
    {
        type      zeroGradient;
    }

    frontAndBack
    {
        type      empty;
    }
}
// ***** //
```

kinematic pressure : $\frac{p}{\rho}$

$$\left[\frac{Pa}{kg/m^3} = m^2 \cdot s^{-2} \right]$$

Unit of kinematic pressure.

$$\frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot (\mathbf{u}\mathbf{u}) - \nabla \cdot (\nu \nabla \mathbf{u}) = -\nabla \frac{p}{\rho}$$

Cavity Flow

- Preprocessing: Boundary and Initial Conditions

```
/*----- C++ -----*/
// =====
// \ \ \ \ \ F i e l d       | OpenFOAM: The Open Source CFD Toolbox
//   \ \ \   O p e r a t i o n | Website: https://openfoam.org
//    \ \ \  A n d             | Version: 12
//     \ \ \ M a n i p u l a t i o n
// -----*/
FoamFile
{
    format      ascii;
    class       volScalarField;
    object      p;
}
// ***** //

dimensions      [0 2 -2 0 0 0 0];

internalField    uniform 0;

boundaryField
{
    movingWall
    {
        type      zeroGradient;
    }

    fixedWalls
    {
        type      zeroGradient;
    }

    frontAndBack
    {
        type      empty;
    }
}
// ***** //
```

In N-S equation, gradient of kinematic pressure is important such that we give a reference value 0 as an uniform internalField.

$$\frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot (\mathbf{u}\mathbf{u}) - \nabla \cdot (\nu \nabla \mathbf{u}) = -\nabla \frac{p}{\rho}$$

Cavity Flow

● Preprocessing: Boundary and Initial Conditions

```
/*----- C++ -----*/
//
// \      /
//  \    /
//   \  /
//    \/
//
Field
Operation
And
Manipulation

OpenFOAM: The Open Source CFD Toolbox
Website: https://openfoam.org
Version: 12

FoamFile
{
    format        ascii;
    class         volScalarField;
    object        p;
}
// *****

dimensions       [0 2 -2 0 0 0 0];

internalField     uniform 0;

boundaryField
{
    movingWall
    {
        type          zeroGradient;
    }
    fixedWalls
    {
        type          zeroGradient;
    }
    frontAndBack
    {
        type          empty;
    }
}

// *****
```

$$\nabla \frac{p}{\rho} = 0 \text{ at the wall}$$

Cavity Flow

- Preprocessing: Physical properties

$$\frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot (\mathbf{u}\mathbf{u}) - \nabla \cdot (\underbrace{\nu}_{\text{Kinematic viscosity}} \nabla \mathbf{u}) = -\nabla \frac{p}{\rho}$$

Kinematic viscosity

Command Line

:wq

cd ../

cd constant

vi physicalProperties

```
/*----- C++ -----*/
//
//  F i e l d
//  O p e r a t i o n
//  A n d
//  M a n i p u l a t i o n
//
//  OpenFOAM: The Open Source CFD Toolbox
//  Website:  https://openfoam.org
//  Version:  12
//
FoamFile
{
    format      ascii;
    class       dictionary;
    location    "constant";
    object      physicalProperties;
}
// *****

viscosityModel constant;
nu              0.01 [m^2/s];
// *****
```

$\nu \text{ [m}^2\text{/s]} = 0.01$

Cavity Flow

- **Preprocessing: Simulation control**

- In the *system* directory,
 - **controlDict**: input data relating to the control of time and reading and writing of the solution
 - fvSchemes, fvSolution: Settings for numerical discretization and solution methods

Command Line

:wq

```
cd ../
```

```
cd system
```

vi controlDict

```

/*-----* C++ *-----*/
FoamFile
{
    format      ascii;
    class       dictionary;
    location    "system";
    object      controlDict;
}

// *****

application      foamRun;

solver            incompressibleFluid;

startFrom        startTime;

startTime        0;

stopAt           endTime;

endTime          10;

deltaT           0.005;

writeControl      timeStep;

writeInterval     100;

purgeWrite        0;

writeFormat       ascii;

writePrecision    6;

writeCompression  off;

timeFormat        general;

timePrecision     6;

runTimeModifiable true;

// *****

```

Cavity Flow

- **Preprocessing: Mesh Generation**
 - The standard mesh generator utility: blockMesh

Command Line

```
cd Tutorial/01-cavity
```

```
blockMesh
```

```
ghkrk@20225263-p01:~/OpenFOAM/ghkrk-12/run/KOSCO-2026-NextGen-Combustion$ cd Tutorial/01-cavity/
ghkrk@20225263-p01:~/OpenFOAM/ghkrk-12/run/KOSCO-2026-NextGen-Combustion/Tutorial/01-cavity$ blockMesh
/*-----*/
      =====
      \\\      / F ield
      \\\      / O peration
      \\\      / A nd
      \\\      / M anipulation
      \\\      /
/*-----*/
OpenFOAM: The Open Source CFD Toolbox
Website:  https://openfoam.org
Version:  12

Build   : 12-86e126a7bc4d
Exec    : blockMesh
Date    : Sep 30 2026
Time    : 19:12:52
Host    : "20225263-p01"
PID     : 14875
I/O     : uncollated
Case    : /home/ghkrk/OpenFOAM/ghkrk-12/run/KOSCO-2026-NextGen-Combustion/Tutorial/01-cavity
nProcs  : 1
sigFpe  : Enabling floating point exception trapping (FOAM_SIGFPE).
fileModificationChecking : Monitoring run-time modified files using timeStampMaster (fileModificationSkew 10)
allowSystemOperations : Allowing user-supplied system call operations
```

Cavity Flow

- Solving: Run the simulation
 - Run the simulation after preprocessing

Command Line

```
foamRun
```

```
Courant Number mean: 0.220549 max: 0.849989  
Time = 9.995s
```

```
smoothSolver: Solving for Ux, Initial residual = 1.19036e-09, Final residual = 1.19036e-09, No Iterations 0  
smoothSolver: Solving for Uy, Initial residual = 2.15087e-08, Final residual = 9.86647e-09, No Iterations 3  
GAMG: Solving for p, Initial residual = 7.46211e-08, Final residual = 7.54887e-09, No Iterations 3  
time step continuity errors : sum local = 6.36133e-11, global = -1.0042e-19  
GAMG: Solving for p, Initial residual = 4.40064e-08, Final residual = 9.6306e-09, No Iterations 2  
time step continuity errors : sum local = 7.75316e-11, global = -7.02276e-19, cumulative = 1.30438e-17  
ExecutionTime = 1.39304 s ClockTime = 2 s
```

```
Courant Number mean: 0.220549 max: 0.849989  
Time = 10s
```

```
smoothSolver: Solving for Ux, Initial residual = 1.39854e-09, Final residual = 1.39854e-09, No Iterations 0  
smoothSolver: Solving for Uy, Initial residual = 2.73188e-08, Final residual = 9.49519e-09, No Iterations 4  
GAMG: Solving for p, Initial residual = 1.06254e-07, Final residual = 4.98108e-09, No Iterations 4  
time step continuity errors : sum local = 4.19039e-11, global = -7.10714e-19  
GAMG: Solving for p, Initial residual = 7.57167e-08, Final residual = 7.6931e-09, No Iterations 3  
time step continuity errors : sum local = 6.32949e-11, global = 2.41785e-19, cumulative = 1.32856e-17  
ExecutionTime = 1.39434 s ClockTime = 2 s
```

```
End
```

Cavity Flow

- Solving: Run the simulation

- You can see the calculated time directories

Command Line

ll

```
ghkrk@20225263-p01:~/OpenFOAM/ghkrk-12/run/KOSCO-2026-NextGen-Combustion/Tutorial/01-cavity$ ll
total 100
drwxr-xr-x 25 ghkrk ghkrk 4096 Oct  1 13:01 ./
drwxr-xr-x  5 ghkrk ghkrk 4096 Oct  1 12:56 ../
drwxr-xr-x  2 ghkrk ghkrk 4096 Oct  1 12:56 0/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  1 13:01 0.5/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  1 13:01 1/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  1 13:01 1.5/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  1 13:01 10/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  1 13:01 2/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  1 13:01 2.5/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  1 13:01 3/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  1 13:01 3.5/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  1 13:01 4/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  1 13:01 4.5/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  1 13:01 5/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  1 13:01 5.5/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  1 13:01 6/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  1 13:01 6.5/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  1 13:01 7/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  1 13:01 7.5/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  1 13:01 8/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  1 13:01 8.5/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  1 13:01 9/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  1 13:01 9.5/
drwxr-xr-x  3 ghkrk ghkrk 4096 Oct  1 13:01 constant/
drwxr-xr-x  2 ghkrk ghkrk 4096 Oct  1 12:56 system/
```

Saved results

Cavity Flow

● Post-Processing

- make dummy file to use Paraview

Command Line

```
touch cavity.foam
```

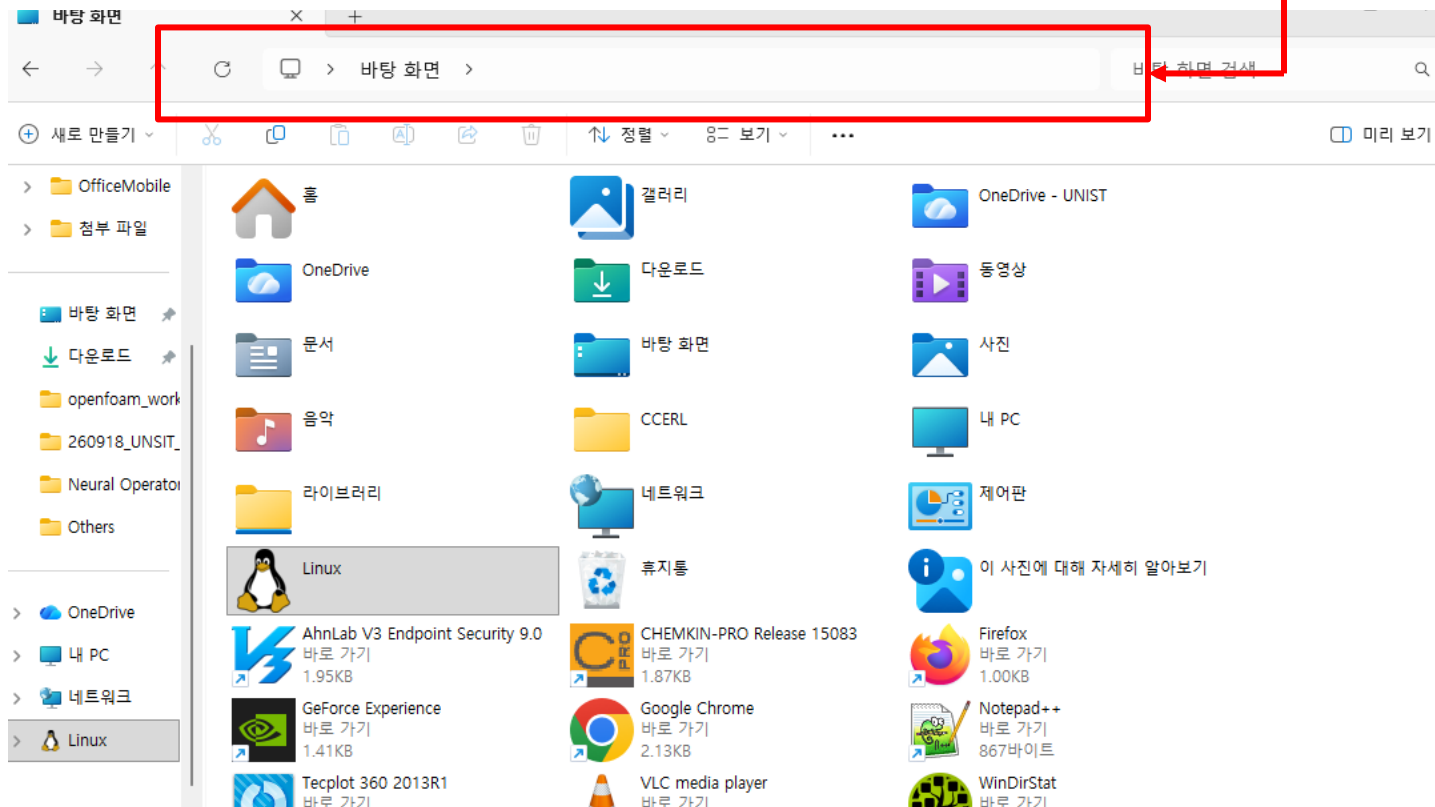
```
ghkrk@20225263-p01:~/OpenFOAM/ghkrk-12/run/KOSCO-2026-NextGen-Combustion/Tutorial/01-cavity$ touch cavity.foam
ghkrk@20225263-p01:~/OpenFOAM/ghkrk-12/run/KOSCO-2026-NextGen-Combustion/Tutorial/01-cavity$ ll
total 1480
drwxr-xr-x 25 ghkrk ghkrk 4096 Oct 1 13:27 ./
drwxr-xr-x 5 ghkrk ghkrk 4096 Oct 1 12:56 ../
drwxr-xr-x 2 ghkrk ghkrk 4096 Oct 1 12:56 0/
drwxr-xr-x 3 ghkrk ghkrk 4096 Oct 1 13:27 0.5/
drwxr-xr-x 3 ghkrk ghkrk 4096 Oct 1 13:27 1/
drwxr-xr-x 3 ghkrk ghkrk 4096 Oct 1 13:27 1.5/
drwxr-xr-x 3 ghkrk ghkrk 4096 Oct 1 13:27 10/
drwxr-xr-x 3 ghkrk ghkrk 4096 Oct 1 13:27 2/
drwxr-xr-x 3 ghkrk ghkrk 4096 Oct 1 13:27 2.5/
drwxr-xr-x 3 ghkrk ghkrk 4096 Oct 1 13:27 3/
drwxr-xr-x 3 ghkrk ghkrk 4096 Oct 1 13:27 3.5/
drwxr-xr-x 3 ghkrk ghkrk 4096 Oct 1 13:27 4/
drwxr-xr-x 3 ghkrk ghkrk 4096 Oct 1 13:27 4.5/
drwxr-xr-x 3 ghkrk ghkrk 4096 Oct 1 13:27 5/
drwxr-xr-x 3 ghkrk ghkrk 4096 Oct 1 13:27 5.5/
drwxr-xr-x 3 ghkrk ghkrk 4096 Oct 1 13:27 6/
drwxr-xr-x 3 ghkrk ghkrk 4096 Oct 1 13:27 6.5/
drwxr-xr-x 3 ghkrk ghkrk 4096 Oct 1 13:27 7/
drwxr-xr-x 3 ghkrk ghkrk 4096 Oct 1 13:27 7.5/
drwxr-xr-x 3 ghkrk ghkrk 4096 Oct 1 13:27 8/
drwxr-xr-x 3 ghkrk ghkrk 4096 Oct 1 13:27 8.5/
drwxr-xr-x 3 ghkrk ghkrk 4096 Oct 1 13:27 9/
drwxr-xr-x 3 ghkrk ghkrk 4096 Oct 1 13:27 9.5/
-rwxr-xr-x 1 ghkrk ghkrk 257 Oct 1 13:25 Allclean*
-rwxr-xr-x 1 ghkrk ghkrk 326 Oct 1 13:04 Allrun*
-rw-r--r-- 1 ghkrk ghkrk 0 Oct 1 13:27 cavity.foam
drwxr-xr-x 3 ghkrk ghkrk 4096 Oct 1 13:27 constant/
-rw-r--r-- 1 ghkrk ghkrk 2052 Oct 1 13:27 log.blockMesh
-rw-r--r-- 1 ghkrk ghkrk 1398949 Oct 1 13:27 log.foamRun
drwxr-xr-x 2 ghkrk ghkrk 4096 Oct 1 13:05 system/
```

Cavity Flow

● Post-Processing

- Find your WSL directory

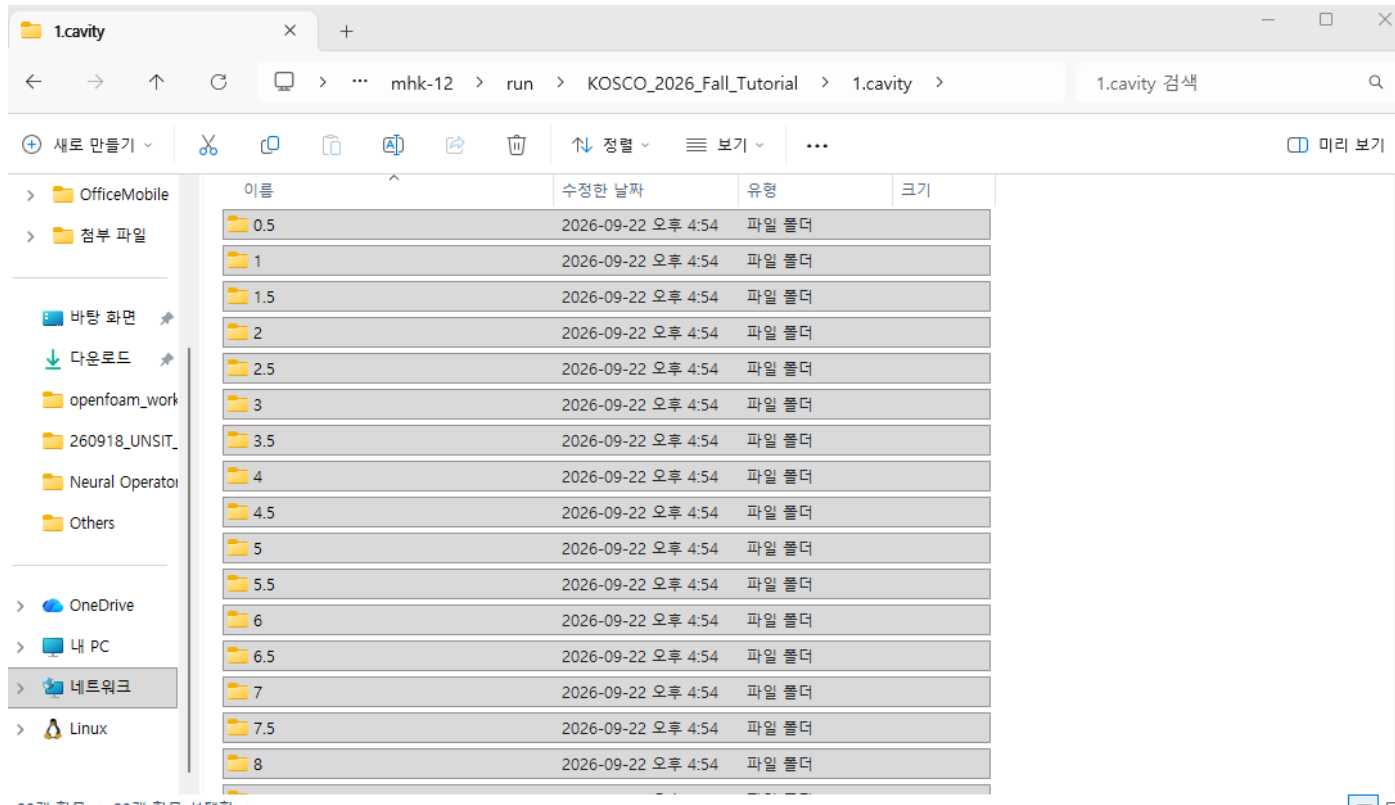
```
\\wsl.localhost\Ubuntu-22.04\home\Username\OpenFOAM\Username-12\run\KOSCO_2026_NextGen-Combustion\1.cavity
```



Cavity Flow

● Post-Processing

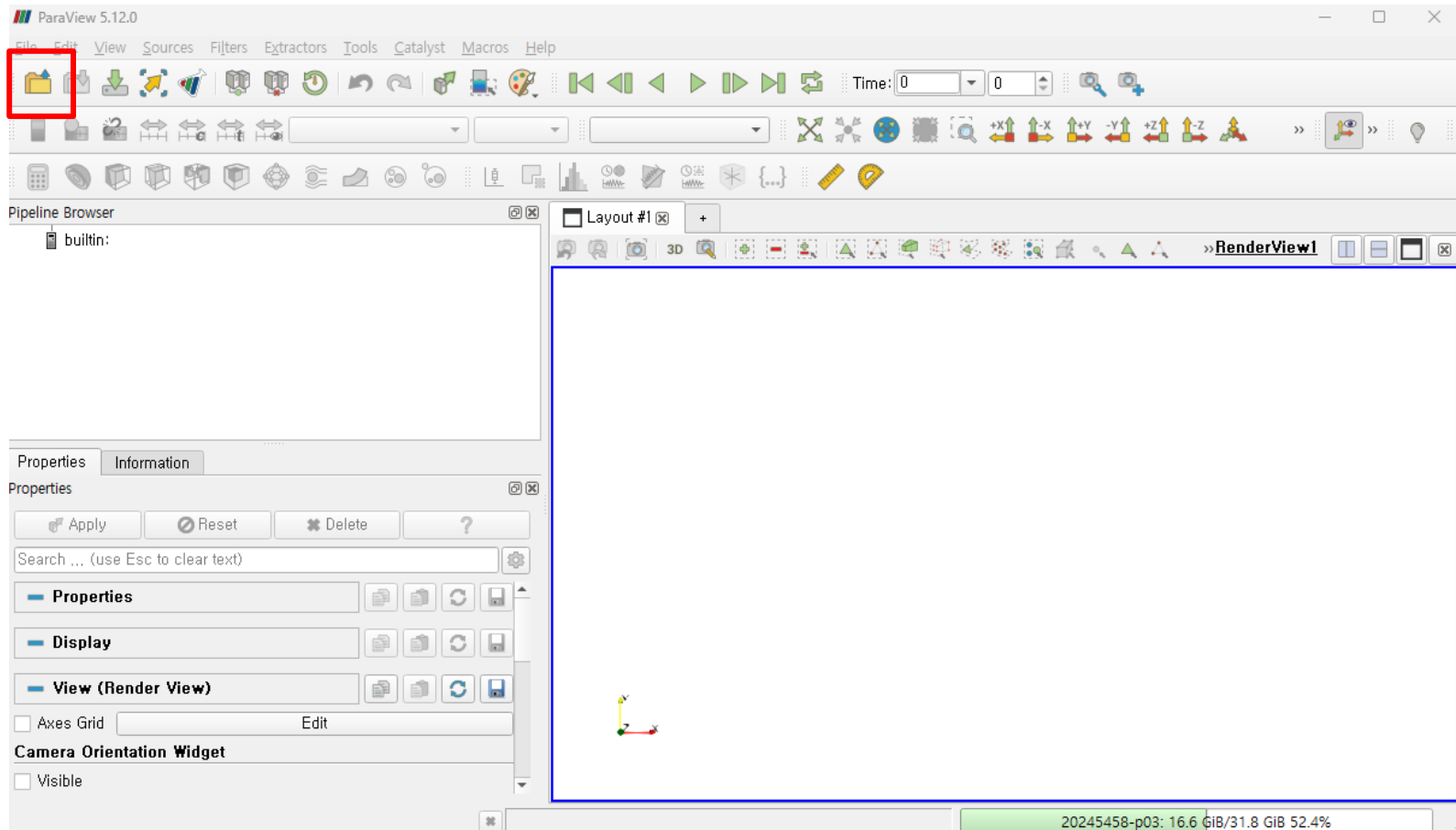
- Copy and paste the whole files anywhere under C: drive or D: drive



Cavity Flow

- **Post-Processing**

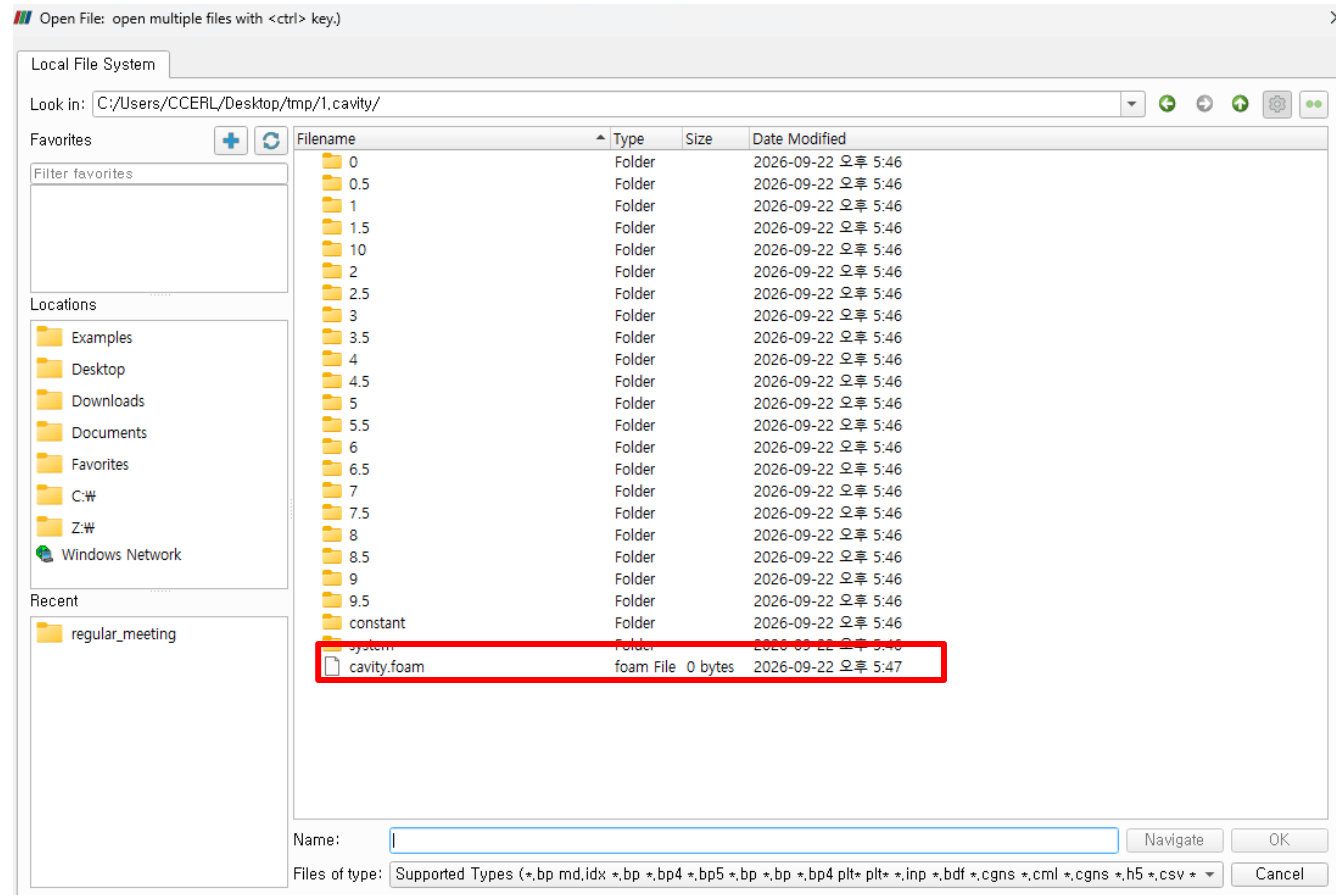
- Open or Browse file



Cavity Flow

- Post-Processing

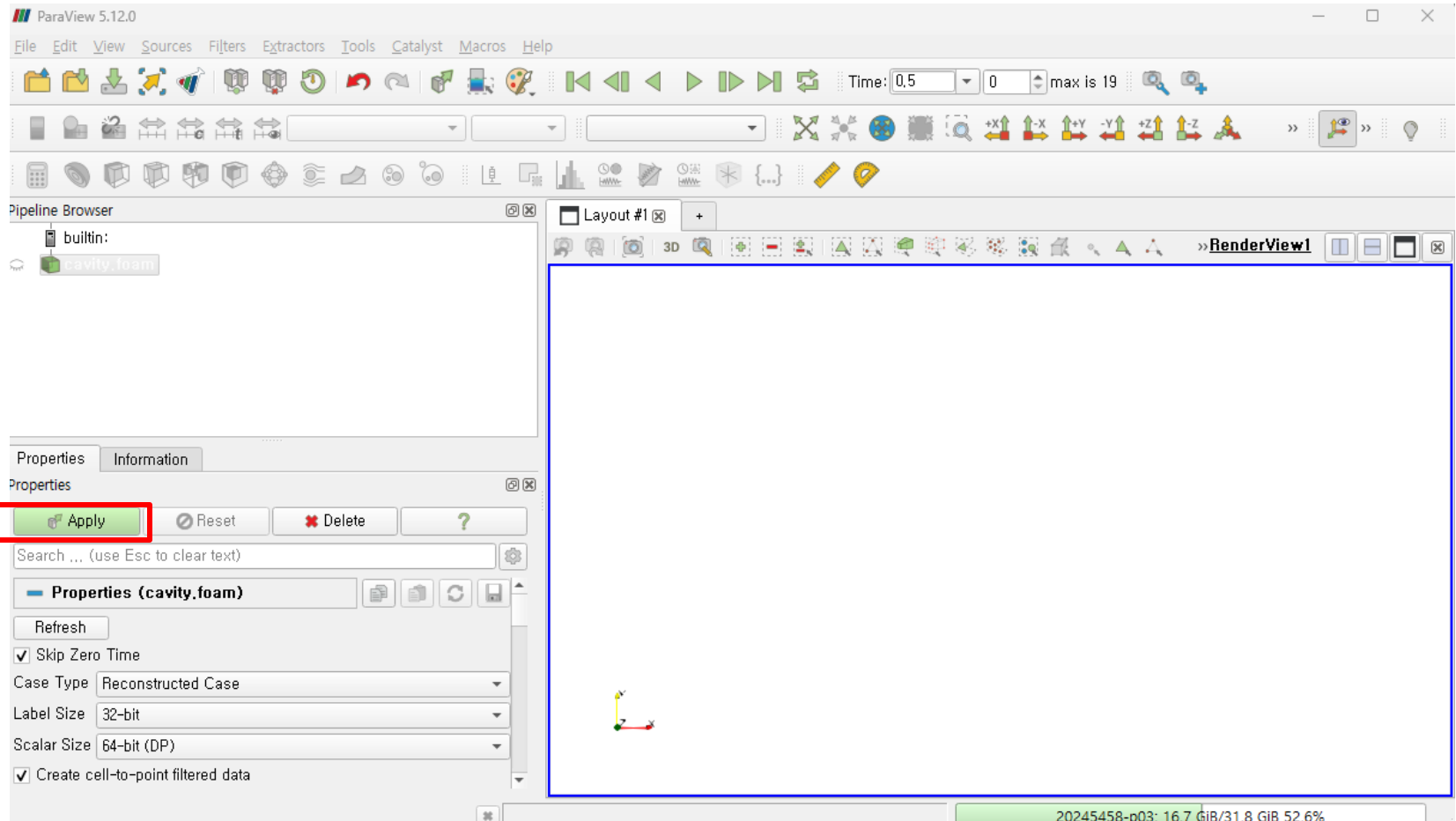
- Choose dummy file (.foam file)



OpenFOAM

- **Post-Processing**

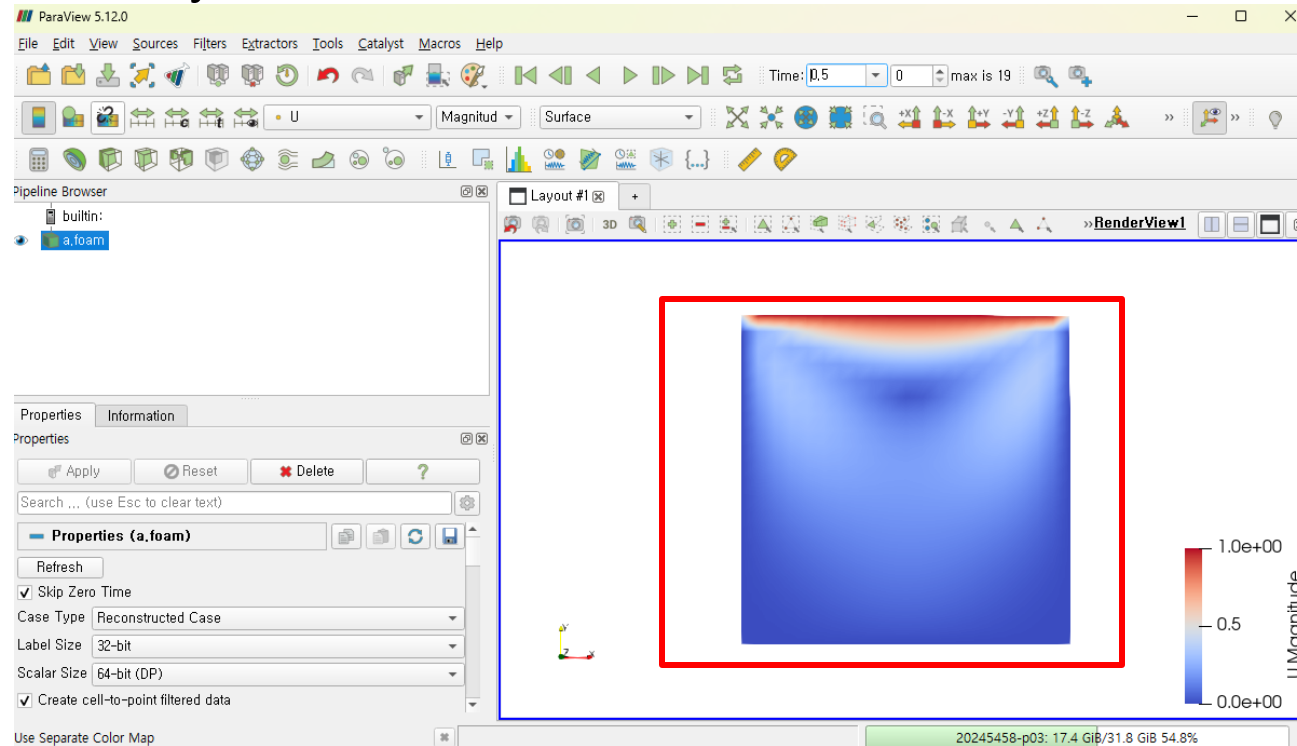
- Press “Apply” to display



Cavity Flow

- **Post-Processing**

- Visualize your simulation



Left click (Ctrl + wheel click/Shift + wheel click) : Change viewpoint

Wheel click (Shift + right click) : move object

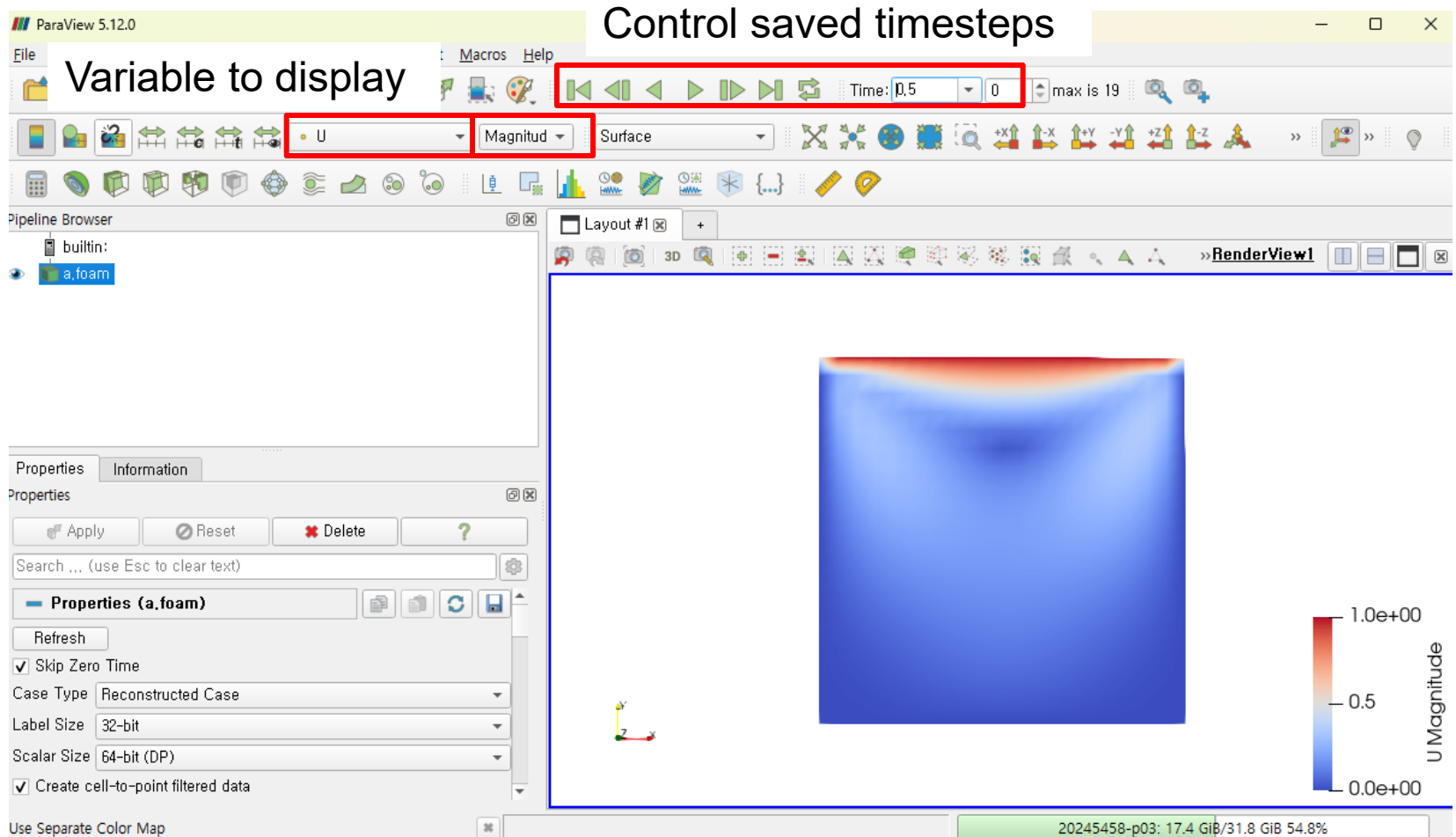
Right click (wheel up,down/Ctrl + right click) : Zoom in, out

Shift + left click : rotate object

Cavity Flow

- **Post-Processing**

- Visualize your simulation



Cavity Flow

- **Post-Processing**

- Visualize your simulation

